

Question 2

Elevator Simulator

PAC

Problem Analysis Chart

Given Data

- N number of floors requests
- Elevator starting floor at 0
- Each floor request

Required Results

- Display elevator movement status for each request
- Update current floor after each request.

Process Required Alternative Solutions

- | | |
|--|--------------------------|
| • Take N as input from the user | Use of Nested if |
| • Take floor request inside while loop for N times | Use of switch statement |
| • Compare request and current floor | Use of else if statement |
| • Display movement status according to the condition | |
| • Set current floor = request | |

elevator simulator

Date: _____

IPO - Input Process Output Chart

Input	Process	Output
Number of requests N	Declare N, i , request, current floor as integ -ex	Display elevator movement status for every request
input for requested floor	set current floor -s and $i \leq 0$	
	Take floor request -st inside while for N times	
	Compare request and current floor	
	Display movement status according to condition	
	set current floor = request.	

Algorithm

1. Start
2. Take input N from the user
3. Set current floor $\leftarrow 0$
4. Set $i \leftarrow 0$
5. while $N > i$ do the following
 - Take input request from the user.
 - if request $>$ current floor do,
 - Display "Moving Up".
 - elseif request $<$ current floor do,
 - Display "Moving down".
 - else do the following
 - Display "Doors opening".
 - Set current floor $\leftarrow$ request
 - increment i by 1
6. Stop

Pseudocode

multitopia

- Begin
- Declare N , request, current floor and i as integer
- Set $i \leftarrow 0$
- Set current floor $\leftarrow 0$
- Input N
- while ($N > i$)
 - Input request
 - if (request > current floor)
 - Output "Moving Up"
 - else if (request < current floor)
 - Output "Moving down"
 - else
 - Output "Doors Opening"
 - end if
 - current floor = request
 - $i = i + 1$
- end while
- End

Flowchart

